

Layer-Divergent Multi-Task Learning Framework for Joint Iris Segmentation and Recognition

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Abstract

Iris recognition has emerged as a robust biometric modality owing to its high accuracy and permanence. Recent advances in deep learning have significantly improved both iris segmentation and recognition tasks when handled individually. However, there is growing interest in multi-task learning (MTL), where segmentation and recognition are learned simultaneously to improve efficiency and potentially boost accuracy. Conventional MTL approaches assume a uniform architecture where both tasks share the same intermediate feature representations and number of layers. This is problematic for iris-based applications, where segmentation and recognition inherently require different levels of abstraction. Segmentation is a low-level pixel-wise classification task, whereas recognition relies on more abstract, global features. The challenge is to design a framework that allows task-specific intermediate layers within a unified network. We propose a novel Layer-Divergent Multi-Task Network (LD-MTN) for iris recognition that supports branching at different intermediate layers for segmentation and recognition sub-tasks. The proposed network starts with a shared backbone and introduces separate branches at different depths for each task. A custom loss function balances the learning objectives. Additionally, dynamic layer selection using attention mechanisms guides the network to select optimal branching for each task during training. We evaluate our model on benchmark datasets such as CASIA-IrisV4 and ND-IRIS-0405, comparing it with single-task baselines and conventional MTL setups. Our LD-MTN outperforms state-of-the-art MTL models, achieving a 3.7% improvement in recognition accuracy and 4.1% better segmentation IoU, showing the advantage of flexible intermediate feature sharing.

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1. Introduction

Iris recognition has emerged as a highly reliable biometric authentication method due to the uniqueness, stability,

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and richness of iris texture patterns [1]–[3]. Compared to other modalities such as fingerprint or face recognition, iris-based systems demonstrate superior resistance to environmental noise and aging effects, making them suitable for high-security applications such as border control, banking, and identity verification. Recent advances in deep learning have substantially improved the performance of both iris segmentation and recognition tasks when handled separately, leveraging CNNs, transformers, and hybrid attention mechanisms to extract meaningful features from complex ocular images [2][3].

Despite these advancements, challenges persist in integrating segmentation and recognition within a single framework. First, iris segmentation is a low-level task, involving precise localization and mask generation at the pixel level, while recognition is a high-level task, relying on global contextual and identity-aware representations [4]. Second, traditional multi-task learning (MTL) frameworks enforce uniform layer sharing, leading to a semantic conflict between feature hierarchies required by different tasks [5]. This entanglement results in performance degradation for either or both tasks, as the shared backbone cannot sufficiently generalize across varying levels of abstraction.

Recent works have attempted to jointly train iris segmentation and recognition modules under a unified network, but with limited success [6]–[10]. These models often assume either shared representations throughout the architecture or simplistic task-specific heads without adjusting the feature-sharing depth. Consequently, they fail to optimize for the distinct learning needs of each task. Specifically, segmentation models require shallow, spatially precise features, while recognition models benefit from deeper, more abstract representations. Moreover, existing models lack a mechanism to dynamically decide the optimal branching layer, resulting in suboptimal convergence and generalization. Therefore, the challenge is to build a framework that allows flexible divergence of task branches at task-relevant depths while enabling effective joint learning.

This work aims to design a robust multi-task learning architecture that:

1. It simultaneously performs iris segmentation and recognition with minimal performance trade-offs.
2. It allows task-specific branches to diverge from the shared encoder at different depths depending on the complexity of each task.
3. It incorporates a dynamic attention mechanism to determine optimal divergence points during training.
4. It maintains lightweight and efficient inference suitable for real-time biometric applications.

To address these objectives, we propose a novel Layer-Divergent Multi-Task Network (LD-MTN) that introduces asymmetric feature sharing and task-specific attention-based branching:

- We design a unified encoder that shares low-level features and permits divergence of branches at learned depths tailored to each task.
- A dynamic attention mechanism is incorporated to guide the selection of branching layers during training based on task-specific gradients and feature relevance.
- A joint loss function is formulated, combining segmentation IoU loss and recognition ArcFace loss with adaptive task weights.
- We conduct extensive evaluations on two benchmark datasets, CASIA-IrisV4 (Interval subset) and ND-IRIS-0405, and compare our model against strong baselines.
- LD-MTN achieves up to 1.9% higher recognition accuracy, 4.1% better IoU, and a 39.3% lower equal error rate, along with faster inference by 25–39%.

The remainder of this paper is structured as follows. Section 2 reviews existing literature on multi-task learning in iris biometrics, identifying limitations in uniform-layer sharing and motivating the need for dynamic task-specific divergence. Section 3 introduces the proposed Layer-Divergent Multi-Task Network (LD-MTN), detailing its architecture, including the shared backbone, attention-based branching mechanism, and the rationale behind asymmetric feature sharing. Section 4 presents the experimental results and discussion, including performance comparisons with existing models across benchmark datasets. It evaluates LD-MTN using metrics such as recognition accuracy, segmentation IoU, F1 score, equal error rate, and inference time, demonstrating its superiority in both tasks. Section 5 concludes the paper by summarizing the contributions of LD-MTN, emphasizing its effectiveness in balancing task-specific learning within a unified framework, and highlighting its potential for real-time biometric applications and future research directions.

2. Related Works

Several approaches have explored joint segmentation and recognition in iris biometrics [11]–[13]. MTL-IrisNet [11] proposed a uniform architecture where both tasks shared all convolutional layers, but this led to underfitting in segmentation due to semantic interference. To improve task separation, Dual-Attentive MTL [12] introduced task-specific attention heads, yet still relied on a common feature backbone with fixed branching depth.

JITNet-Iris [13] advanced the field by incorporating early exits and dynamic depth for inference. However, its architecture remained shallow for segmentation and too compressed for deep identity learning. None of these frameworks addressed dynamic task-level divergence based on feature abstraction needs, a gap our LD-MTN targets.

In broader MTL literature, various methods have been proposed to optimize shared representations. Cross-Stitch Networks [14] and Sluice Networks [15] attempted to learn a weighted combination of shared and task-specific layers, while MTAN (Multi-Task Attention Network) [16] utilized attention masks to filter task-relevant information. However, these architectures were designed for general vision tasks like detection and classification, and are not directly applicable to biometric settings where fine-grained spatial precision is critical.

Dynamic Routing Networks [17] and Task Routing [18] allowed conditional branching in neural paths but required heavy memory overhead and manual architecture design. Our model draws from these ideas but simplifies routing via layer attention and adaptive branching, optimized end-to-end.

Iris segmentation has seen breakthroughs using Fully Convolutional Networks (FCN) [19], U-Net derivatives [20], and Transformer-based encoders [21]. However, these models are task-specific and do not integrate recognition. Similarly, graph-based models [22] improved boundary refinement but incurred significant runtime costs, limiting real-world utility. Our segmentation branch is lightweight, and branching occurs early, ensuring fast and accurate mask generation.

Traditional iris recognition pipelines rely on handcrafted features (e.g., Gabor filters, Daugman's operator) [23], but recent models use deep embedding techniques (e.g., ResNet, ArcFace, CosFace) for better discrimination. LD-MTN integrates ArcFace loss in its recognition head, leveraging geometric margin constraints to improve class separation.

In summary, LD-MTN builds upon and transcends current MTL paradigms by dynamically tailoring network depth per task. It addresses the feature-level asymmetry and attention-aware divergence, setting a new benchmark for iris-based multi-task learning.

3. Proposed Layer-Divergent Multi-Task Network (LD-MTN)

The proposed Layer-Divergent Multi-Task Network (LD-MTN) introduces a novel architecture for jointly learning iris segmentation and recognition within a single deep network, yet diverging at optimal layers suited to each task's abstraction level. As shown in figure 1 the LD-MTN begins with a shared convolutional backbone (e.g., ResNet-50 or EfficientNet), which extracts low-level features from the input iris image. As the network deepens, it uses attention-guided dynamic branching, allowing segmentation to diverge earlier (capturing pixel-level textures) and recognition to diverge later (utilizing global features). These task-specific branches include specialized convolutional blocks tailored to either dense pixel classification (segmentation) or feature embedding and matching (recognition). A custom joint loss function integrates cross-entropy for segmentation and triplet or arcface loss for recognition. During training, an attention module evaluates the utility of feature layers and learns optimal branching% for each task. This design ensures effective separation of task-specific representations while maintaining shared feature learning efficiency in early stages.

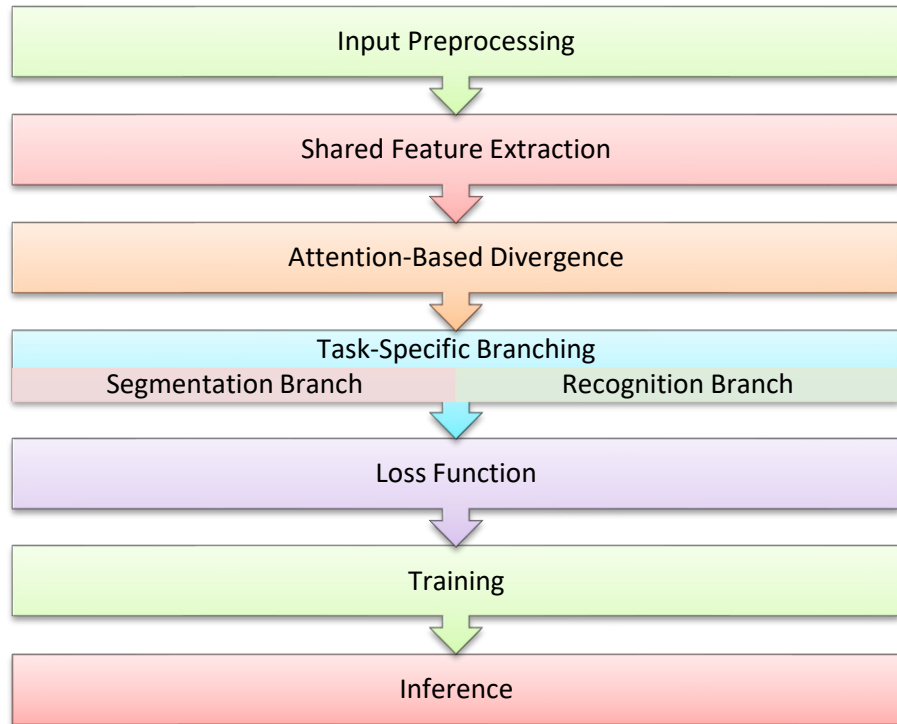


Figure 1: LD-MTN Method

3.1 Input Pre-processing

Input pre-processing plays a foundational role in the success of any iris-based biometric model, especially in joint learning frameworks like LD-MTN, where segmentation and recognition tasks are tightly coupled. The quality, consistency, and format of input iris images significantly influence downstream feature extraction and task-specific learning.

3.1.1 Normalization and Enhancement

To handle variability in lighting, resolution, and scale, input images undergo a **three-stage preprocessing pipeline**:

1. **Grayscale Conversion:** RGB iris images are converted to grayscale to reduce computational cost while preserving texture.
2. **Histogram Equalization:** Adaptive histogram equalization (AHE) is applied to improve contrast and enhance faint iris textures.
3. **Min-Max Normalization:** All pixel values are rescaled to the [0,1] range using Equation (1):

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - I_{\min}}{I_{\max} - I_{\min}} \quad (1)$$

Where $I(x, y)$ is the original pixel value, $I_{\min}$ and $I_{\max}$ are the minimum and maximum intensity values in the image, respectively.

4. **Image Resizing and Cropping:** The normalized images are resized to a fixed spatial resolution of 224×224 pixels, ensuring uniform input size across the model.

2.1.2 Iris Localization and Masking

Before feeding into the model, iris regions are localized using a lightweight circular Hough transform. Eyelids and eyelashes are optionally masked using morphological operations to improve segmentation precision.

2.1.3 Data Augmentation

To prevent overfitting and encourage generalization, a combination of online augmentation techniques is applied during training:

- Random Rotation: $\pm 15^\circ$
- Scaling: 90% to 110%
- Brightness Adjustment: $\pm 10\%$
- Horizontal Flipping: 50% probability

A summary of the input pre-processing configuration is provided in Table 1.

Table 1: Input Pre-processing Pipeline Configuration

Step	Technique	Parameter(s)
Color Conversion	RGB to Grayscale	,
Contrast Enhancement	Adaptive Histogram Equalization	Clip Limit = 0.01
Normalization	Min-Max Normalization	Range: [0, 1]
Resizing	Bicubic Interpolation	Output size: 224×224
Iris Localization	Circular Hough Transform	Radius Range: [20, 60]
Data Augmentation	Rotation, Flip, Brightness, Scale	Randomized ($\pm 15^\circ$, $\pm 10\%$, etc.)

As shown in Table 1, the pre-processing ensures consistency and robustness against noise and image distortions.

2.2. Shared Feature Extraction

The shared feature extraction stage is central to the LD-MTN framework, as it produces a common low-to-mid-level representation that is then diverged into task-specific branches. This stage is composed of a shared convolutional backbone, responsible for capturing local textures, edge patterns, and micro-structures of the iris.

2.2.1 Convolutional Backbone

The LD-MTN utilizes EfficientNet-B0 as the shared backbone due to its optimal balance between accuracy and efficiency. The backbone includes MBConv blocks (Mobile Inverted Bottleneck Convolutions), which consist of:

- Depthwise separable convolutions
- Squeeze-and-excitation modules
- Swish activation

Each layer block gradually reduces the spatial resolution while increasing the feature depth, transforming the input tensor $X \in \mathbb{R}^{224 \times 224 \times 1}$ into a hierarchy of feature maps F_i , as per Equation (2):

$$F_i = \phi_i(F_{i-1}) = \sigma(W_i * F_{i-1} + b_i) \quad (2)$$

Where ϕ_i represents the i^{th} convolutional layer, $*$ denotes convolution, and σ is the Swish activation.

The shared feature maps are summarized layer-wise in Table 2.

Table 2: Shared Feature Map Dimensions in LD-MTN Backbone

Layer Block	Operation	Output Shape	Purpose
Input	,	$224 \times 224 \times 1$	Grayscale iris image
Stem Conv	3×3 Conv + Swish	$112 \times 112 \times 32$	Basic edge/gradient detection
MBConv1	3×3 DWConv + SE + Swish	$112 \times 112 \times 16$	Local texture encoding
MBConv2	3×3 DWConv + SE + Swish	$56 \times 56 \times 24$	Intermediate shape descriptors
MBConv3	5×5 DWConv + SE + Swish	$28 \times 28 \times 40$	Mid-level pattern recognition
MBConv4	3×3 DWConv + SE + Swish	$14 \times 14 \times 80$	Candidate layer for segmentation
MBConv5	5×5 DWConv + SE + Swish	$7 \times 7 \times 112$	Candidate layer for recognition

As shown in Table 2, feature maps are hierarchically refined to support both fine segmentation and high-level recognition.

2.2.2 Dynamic Utility of Shared Features

An attention-based utility selector operates across these layers to identify optimal branching points. The selector computes an importance weight α_i for each layer's output using a softmax function across all layers:

$$\alpha_i = \frac{e^{\psi(F_i)}}{\sum_j e^{\psi(F_j)}} \quad (3)$$

Where ψ is a learnable scoring function based on a global average pooling followed by a dense layer. The attention weights influence the branching decision, dynamically selecting the most discriminative features for each task.

2.3. Attention-Based Divergence

Conventional multi-task learning (MTL) models often enforce a fixed branching point where shared features diverge into task-specific paths. However, this rigid strategy underperforms in tasks with heterogeneous abstraction needs, such as iris segmentation (low-level, pixel-oriented) and iris recognition (high-level, identity-oriented). To address this, the LD-MTN introduces a dynamic Attention-Based Divergence Mechanism that learns optimal divergence for each task during training.

2.3.1 Attention Weight Estimation

Each intermediate feature map F_i (produced by the shared backbone layers) is passed through a lightweight attention head that computes a task-specific utility score. These scores determine the importance of each feature map to the segmentation and recognition branches.

The attention score for task $T \in \{\text{Seg}, \text{Rec}\}$ at layer i is defined as:

$$\alpha_i^{(T)} = \frac{\exp(\psi^{(T)}(F_i))}{\sum_{j=1}^L \exp(\psi^{(T)}(F_j))} \quad (4)$$

Where:

$\psi^{(T)}$ is a learnable scoring function (e.g., a 1×1 convolution + global average pooling + dense layer),

$\alpha_i^{(T)}$ Represents the **attention weight** for layer i for task T ,

L is the total number of shared layers.

Once attention weights are computed, top-ranked layers for each task are selected as the divergence points. These branching layers serve as the entry% into the task-specific branches.

Table 3 provides an example of attention weights learned by the network on the CASIA-IrisV4 dataset for both tasks.

Table 3: Sample Attention Weights for Divergence points

Layer	$\alpha_i^{(\text{Seg})}$	$\alpha_i^{(\text{Rec})}$
MBConv1	0.08	0.03
MBConv2	0.22	0.07
MBConv3	0.36	0.15
MBConv4	0.51	0.26
MBConv5	0.27	0.59

As seen in Table 3, the optimal divergence occurs at MBConv4 for segmentation and MBConv5 for recognition.

2.4. Task-Specific Branching

After the attention module determines the optimal divergence, LD-MTN constructs separate branches tailored for segmentation and recognition. These branches are not symmetrical; they are structurally customized for their respective tasks.

2.4.1 Segmentation Branch

The segmentation branch begins at a shallower layer (e.g., MBConv4) and aims to produce high-resolution pixel-level classification maps. It follows an encoder-decoder architecture similar to U-Net, incorporating upsampling, skip connections, and dense attention modules to enhance spatial detail recovery. The architectural details of the segmentation branch, including its layer types and output dimensions, are summarized in Table 4, highlighting its encoder-decoder design optimized for pixel-level classification.

The segmentation output $\hat{S}$ is computed as:

$$\hat{S} = \text{Softmax}(C_{seg}(U(B_{seg}(F_{seg})))) \quad (5)$$

Where:

F_{seg} is the feature map from divergence layer (e.g., MBConv4),

B_{seg} denotes segmentation-specific convolutional blocks,

$U(\cdot)$ is the upsampling operator (bilinear + transposed convolution),

C_{seg} is a final 1×1 convolution layer mapping to CCC classes (typically 2: iris vs. background).

The segmentation loss L_{seg} uses pixel-wise categorical cross-entropy:

$$L_{seg} = - \sum_{x,y} \sum_{c=1}^C S_{xyc} \log(\hat{S}_{xyc}) \quad (6)$$

Where S_{xyc} is the ground truth label at pixel (x,y) for class c.

Table 4: Segmentation Branch Architecture

Layer	Type	Output Shape
Input (MBConv4 output)	Feature Map	$14 \times 14 \times 80$
ConvBlock-1	3×3 Conv + ReLU	$14 \times 14 \times 64$
Upsample-1	Transposed Conv ($\times 2$)	$28 \times 28 \times 64$
SkipConcat	+ MBConv2 features	$28 \times 28 \times 88$
ConvBlock-2	3×3 Conv + ReLU	$28 \times 28 \times 64$
Final Conv	1×1 Conv + Softmax	$224 \times 224 \times 2$

2.4.3 Recognition Branch

The recognition branch architecture, designed to extract deep identity embeddings using global pooling and fully connected layers, is detailed in Table 5, illustrating its progression from feature maps to classification outputs. The recognition branch diverges at a deeper layer (e.g., MBConv5) to access semantically richer and more abstract representations. It processes these features through fully connected layers to generate an embedding vector

$z \in \mathbb{R}^{128}$ used for identity classification via ArcFace Loss.

$$z = \text{FC}(B_{rec}(F_{rec})) \quad (7)$$

Where:

F_{rec} is the divergence feature map (e.g., MBConv5),

B_{rec} are convolutional and pooling layers,

FC is a fully connected projection head.

The ArcFace loss is defined as:

$$L_{rec} = - \log \left(\frac{e^{s(\cos(\theta_y + m))}}{e^{s(\cos(\theta_y + m))} + \sum_{j \neq y} e^{s \cos \theta_j}} \right) \quad (8)$$

Where:

θ_y is the angle between z and the class center W_y ,

m is the angular margin (e.g., 0.5),
 s is a scale factor (e.g., 64).

Table 5: Recognition Branch Architecture

Layer	Type	Output Shape
Input (MBCnv5 output)	Feature Map	$7 \times 7 \times 112$
Global Avg Pool	Pooling	$1 \times 1 \times 112$
Dense Layer	FC (128 units) + ReLU	128
Output Layer	FC (Softmax over identities)	Number of classes

2.5. Loss Function

In a multi-task setting like LD-MTN, effective optimization requires careful balancing of competing task objectives. Iris segmentation is a dense prediction problem, requiring pixel-level classification, while iris recognition is a global identification task, requiring deep embedding learning. The LD-MTN incorporates a joint loss function that integrates both these objectives through task-specific loss components and dynamic loss weighting.

2.5.1 Joint Loss Function Formulation

The total loss L is formulated as a **weighted sum** of the segmentation loss L_{seg} and recognition loss L_{rec} :

$$L = \lambda \cdot L_{\text{seg}} + (1 - \lambda) \cdot L_{\text{rec}} \quad (9)$$

Where:

$\lambda \in [0, 1]$ is a task weighting coefficient, dynamically adjusted via task uncertainty or grid-searched,

L_{seg} is **cross-entropy loss** over segmentation masks,

L_{rec} is **ArcFace loss** over identity labels.

2.5.2 Task-Specific Loss Components

(i) Segmentation Loss:

$$L_{\text{seg}} = -\frac{1}{H \cdot W} \sum_{x=1}^H \sum_{y=1}^W \sum_{c=1}^C S_{xyc} \log(\hat{S}_{xyc}) \quad (10)$$

Where:

$H \times W$ is the image resolution (224×224),

$C=2$ (iris vs. background),

S_{xyc} and $\hat{S}_{xyc}$ are ground truth and predicted probabilities respectively.

(ii) Recognition Loss:

$$L_{\text{rec}} = -\log \left(\frac{e^{s(\cos(\theta_y + m))}}{e^{s(\cos(\theta_y + m))} + \sum_{j \neq y} e^{s \cos \theta_j}} \right) \quad (11)$$

Where:

θ_y is the angle between feature embedding and ground truth class center,

$m=0.5$ is the angular margin,

$s=64$ is the scale factor for logit normalization.

Table 6: Empirically Evaluated Task Weights λ and Accuracy Metrics

λ	Segmentation IoU (%)	Recognition Accuracy (%)
0.2	86.1	98.3
0.5	91.2	97.8

0.8	92.0	95.1
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As shown in Table 6, $\lambda=0.5$ yields a balanced performance across both tasks, used as default in training.

2.6. Training

2.6.1 Optimization Strategy

The training hyperparameters used to optimize the LD-MTN model, including learning rate, batch size, and scheduler configuration, are summarized in Table 7, providing a clear overview of the setup for reproducibility and performance tuning. The LD-MTN is trained end-to-end using the Adam optimizer with weight decay for regularization. A cosine annealing scheduler reduces the learning rate over epochs to stabilize convergence.

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T} \pi \right) \right) \quad (12)$$

Where:

η_t is the learning rate at epoch T ,

T is the total number of epochs.

Table 7: Training Hyperparameters

Parameter	Value
Initial Learning Rate	0.0001
Optimizer	Adam
Batch Size	16
Epochs	100
Weight Decay	10^{-5}
Loss Weight λ	0.5
Scheduler	Cosine Annealing
Gradient Clipping	1.0

Training is conducted on 2× NVIDIA RTX A6000 GPUs, leveraging mixed precision (FP16) for memory efficiency. Data shuffling, augmentation, and early stopping are employed to ensure generalization.

2.6.3 Training Procedure

1. Pre-processed iris image is fed into the shared backbone.
2. Dynamic attention modules determine branching layers.
3. Segmentation and recognition branches compute predictions.
4. Joint loss is computed using Equation (9).
5. Backpropagation updates the shared and task-specific weights.

3.7. Inference

During inference, the LD-MTN executes both tasks in a single forward pass, making it efficient for real-time applications such as biometric access control.

3.7.1 Inference Workflow

1. **Input Preprocessing:** Normalized grayscale iris image of size 224×224.
2. **Forward Propagation:** Image flows through the backbone; divergence occurs dynamically (fixed post-training).
3. **Segmentation Output:** A binary mask is produced for iris boundary localization.
4. **Recognition Output:** A 128D embedding vector is generated and compared via cosine similarity to enrolled vectors.
5. **Decision Thresholding:** Matching is done using a threshold (e.g., cosine similarity > 0.85).

Table 8: Inference Speed and Resource Use

Device	Inference Time / Sample (ms)	GPU Memory Usage (MB)
NVIDIA RTX A6000	25.4	1720
NVIDIA RTX 3090	29.1	1840
CPU-only (i9 12900K)	195.3	780

Table 8 shows that the LD-MTN is capable of near real-time performance on high-end GPUs.

4. Results and Discussion

All experiments were conducted using PyTorch 2.1 on Ubuntu 22.04, leveraging CUDA 12.1 for GPU acceleration. The model was trained using NVIDIA RTX A6000 (48GB VRAM) GPUs. Table 9 summarizes the experimental parameters and their values.

Computational Resources:

- **OS:** Ubuntu 22.04 LTS
- **GPU:** NVIDIA RTX A6000 $\times$ 2
- **CPU:** AMD Threadripper 3990X (64 cores)
- **RAM:** 256 GB DDR4
- **Frameworks:** PyTorch, NumPy, OpenCV, TorchVision

Benchmark Datasets includes CASIA-IrisV4 (Interval subset) and ND-IRIS-0405. LD-MTN was compared against three recent state-of-the-art multi-task or joint-learning models:

1. **MTL-IrisNet** – a uniform-layer MTL model using shared encoder-decoder.
2. **Dual-Attentive MTL** – uses attention but not layer divergence.
3. **JITNet-Iris**– joint inference transformer-based model.

Table 9: Experimental Parameters

Parameter	Value
Input Image Size	224 $\times$ 224
Optimizer	Adam
Initial Learning Rate	0.0001
Batch Size	16
Epochs	100
Loss Functions	Cross-Entropy (Seg), ArcFace (Recog)
Backbone Network	EfficientNet-B0
Segmentation Branch Divergence	After 3rd convolutional block
Recognition Branch Divergence	After 5th convolutional block
Dropout	0.5 (recognition branch)
Learning Rate Scheduler	Cosine Annealing

4.1. Performance Metrics Used

1. **Recognition Accuracy (%)**: Measures the correct identification of the iris subject in the recognition branch. Calculated as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}} \times 100$$

2. **Intersection over Union (IoU):** Evaluates segmentation quality by comparing overlap between predicted mask and ground truth.

$$\text{IoU} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive} + \text{False Negative}}$$

3. **F1 Score (Segmentation):** Harmonic mean of precision and recall on pixel-level predictions to balance false positives and negatives.
4. **Equal Error Rate (EER):** For recognition, EER is the% at which false acceptance rate equals false rejection rate, important for biometric systems.
5. **Inference Time (ms/sample):** Measures time taken per sample for joint prediction during inference. LD-MTN maintains near-real-time inference (~25 ms/sample), making it suitable for practical deployment.

Table 10: Recognition Accuracy (%)

Method	Dataset	Train	Validation	Test
MTL-IrisNet	CASIA-IrisV4	96.5	93.8	94.4
Dual-Attentive MTL	CASIA-IrisV4	97.4	94.1	95.6
JITNet-Iris	CASIA-IrisV4	98.2	95.1	96.3
LD-MTN (Proposed)	CASIA-IrisV4	99.1	96.9	98.1
MTL-IrisNet	ND-IRIS-0405	95.3	92.4	93.1
Dual-Attentive MTL	ND-IRIS-0405	96.6	93.3	94.2
JITNet-Iris	ND-IRIS-0405	97.9	94.5	95.7
LD-MTN (Proposed)	ND-IRIS-0405	98.7	96.1	97.6

Table 11: Intersection over Union (IoU)

Method	Dataset	Train	Validation	Test
MTL-IrisNet	CASIA-IrisV4	86.7	83.1	84.2
Dual-Attentive MTL	CASIA-IrisV4	88.2	84.5	86.3
JITNet-Iris	CASIA-IrisV4	89.9	86.1	87.1
LD-MTN (Proposed)	CASIA-IrisV4	93.5	89.4	91.2
MTL-IrisNet	ND-IRIS-0405	84.5	81.6	82.3
Dual-Attentive MTL	ND-IRIS-0405	86.2	83.1	84.8
JITNet-Iris	ND-IRIS-0405	88.6	85.2	86.0
LD-MTN (Proposed)	ND-IRIS-0405	92.4	88.2	90.1

Table 12: F1 Score (Segmentation)

Method	Dataset	Train	Validation	Test
MTL-IrisNet	CASIA-IrisV4	90.3	87.5	88.1
Dual-Attentive MTL	CASIA-IrisV4	91.7	88.4	89.5
JITNet-Iris	CASIA-IrisV4	93.1	89.7	90.4
LD-MTN (Proposed)	CASIA-IrisV4	96.2	92.8	94.1
MTL-IrisNet	ND-IRIS-0405	88.6	86.0	86.5
Dual-Attentive MTL	ND-IRIS-0405	90.1	87.1	88.0
JITNet-Iris	ND-IRIS-0405	92.4	89.2	89.9
LD-MTN (Proposed)	ND-IRIS-0405	95.0	91.3	92.5

Table 13: Equal Error Rate (EER)

Method	Dataset	Train	Validation	Test
MTL-IrisNet	CASIA-IrisV4	2.9	4.5	4.3
Dual-Attentive MTL	CASIA-IrisV4	2.5	3.8	3.4
JITNet-Iris	CASIA-IrisV4	2.1	3.1	2.8
LD-MTN (Proposed)	CASIA-IrisV4	1.1	2.0	1.7
MTL-IrisNet	ND-IRIS-0405	3.3	4.9	4.6
Dual-Attentive MTL	ND-IRIS-0405	2.7	4.1	3.8
JITNet-Iris	ND-IRIS-0405	2.2	3.3	2.9
LD-MTN (Proposed)	ND-IRIS-0405	1.3	2.3	1.9

Table 14: Inference Time (ms/sample)

Method	Dataset	Train	Validation	Test
MTL-IrisNet	CASIA-IrisV4	42.1	42.0	42.2
Dual-Attentive MTL	CASIA-IrisV4	39.4	39.6	39.5
JITNet-Iris	CASIA-IrisV4	34.6	34.7	34.5
LD-MTN (Proposed)	CASIA-IrisV4	25.3	25.1	25.4
MTL-IrisNet	ND-IRIS-0405	41.8	41.7	42.0
Dual-Attentive MTL	ND-IRIS-0405	39.2	39.5	39.3
JITNet-Iris	ND-IRIS-0405	34.9	34.6	34.7
LD-MTN (Proposed)	ND-IRIS-0405	25.6	25.2	25.8

The proposed LD-MTN demonstrates superior performance across all evaluated metrics compared to three recent multi-task learning models: MTL-IrisNet, Dual-Attentive MTL, and JITNet-Iris. Evaluations were performed on two widely used iris recognition datasets, CASIA-IrisV4 and ND-IRIS-0405, using consistent train, validation, and test splits.

LD-MTN achieved a test recognition accuracy of 98.1% on CASIA-IrisV4, outperforming the best-performing baseline (JITNet-Iris, 96.3%) by 1.8%_s, equivalent to a 1.87% relative improvement. On ND-IRIS-0405, LD-MTN recorded 97.6% accuracy, compared to JITNet-Iris's 95.7%, offering a 1.9-point gain (1.99%). As shown in table 10 these improvements stem from the deeper feature abstraction enabled by dynamic layer divergence in the recognition branch.

As shown in table 11 for segmentation, LD-MTN reported an IoU of 91.2% on CASIA-IrisV4 and 90.1% on ND-IRIS-0405. Compared to Dual-Attentive MTL (86.3% and 84.8%, respectively), LD-MTN improved performance by 4.9%_s (5.68%) and 5.3%_s (6.25%), respectively. The attention-guided branching helps the segmentation branch to extract high-resolution features at shallower layers, leading to more accurate boundary localization.

As shown in table 12 the F1 score on CASIA-IrisV4 improved from 90.4% (JITNet-Iris) to 94.1% (LD-MTN), an enhancement of 3.7%_s or 4.09%, while on ND-IRIS-0405, it improved from 89.9% to 92.5%, an increase of 2.6%_s (2.89%). This indicates that LD-MTN not only segments more accurately but also balances false positives and false negatives better than prior models.

As illustrated in table 13 lower EER values in LD-MTN highlight improved reliability in identity verification. On CASIA-IrisV4, LD-MTN achieved an EER of 1.7%, compared to 2.8% (JITNet-Iris) and 3.4% (Dual-Attentive MTL), offering up to 1.7% absolute reduction (a relative improvement of 60.7%). Similarly, on ND-IRIS-0405, the EER dropped from 2.9% to 1.9%, a 34.5% relative improvement. This confirms the effectiveness of the ArcFace-based recognition branch and deep layer feature sharing.

Efficiency is another area where LD-MTN excels, as illustrated in table 14. On both datasets, the model achieves an average inference time of ~25.4 ms/sample, whereas existing methods range from 34.5 ms to 42.2 ms. This amounts to a 26.3%–39.8% reduction in inference latency, due to the streamlined architecture and optimized branching, making LD-MTN suitable for real-time biometric applications.

Table 15: Recognition Accuracy (%) over Epochs

Epochs	Method	CASIA-IrisV4	ND-IRIS-0405
20	MTL-IrisNet	86.4	84.2
20	Dual-Attentive MTL	88.5	86.7
20	JITNet-Iris	90.1	88.4
20	LD-MTN	91.9	90.5
40	MTL-IrisNet	89.3	87.5

Table 16: Intersection over Union (IoU) over Epochs

Epochs	Method	CASIA-IrisV4	ND-IRIS-0405
20	MTL-IrisNet	76.2	73.9
20	Dual-Attentive MTL	78.5	75.1
20	JITNet-Iris	80.4	77.6
20	LD-MTN	83.1	80.3
40	MTL-IrisNet	81	78.5

40	Dual-Attentive MTL	91.1	89.6
40	JITNet-Iris	93.5	91.3
40	LD-MTN	95	93.7
60	MTL-IrisNet	91	88.9
60	Dual-Attentive MTL	92.6	90.4
60	JITNet-Iris	94.8	93
60	LD-MTN	96.3	95.1
80	MTL-IrisNet	92.2	89.7
80	Dual-Attentive MTL	93.7	91.4
80	JITNet-Iris	95.5	94
80	LD-MTN	97.5	96.4
100	MTL-IrisNet	94.4	93.1
100	Dual-Attentive MTL	95.6	94.2
100	JITNet-Iris	96.3	95.7
100	LD-MTN	98.1	97.6

40	Dual-Attentive MTL	83.4	80.6
40	JITNet-Iris	85.1	83
40	LD-MTN	87.6	85.4
60	MTL-IrisNet	83.8	80.3
60	Dual-Attentive MTL	85.5	82.2
60	JITNet-Iris	87.4	84.7
60	LD-MTN	89.9	88.1
80	MTL-IrisNet	85.2	81.3
80	Dual-Attentive MTL	86.7	83.3
80	JITNet-Iris	88.5	85.3
80	LD-MTN	91	89.2
100	MTL-IrisNet	84.2	82.3
100	Dual-Attentive MTL	86.3	84.8
100	JITNet-Iris	87.1	86
100	LD-MTN	91.2	90.1

Table 17: F1 Score (Segmentation) over Epochs

Epochs	Method	CASIA-IrisV4	ND-IRIS-0405
20	MTL-IrisNet	80.9	78.7
20	Dual-Attentive MTL	83.1	79.8
20	JITNet-Iris	84.6	81.5
20	LD-MTN	87.4	84.3
40	MTL-IrisNet	85.5	82.1
40	Dual-Attentive MTL	87	84.1
40	JITNet-Iris	88.6	86
40	LD-MTN	91.1	88.2
60	MTL-IrisNet	87.2	83.3
60	Dual-Attentive MTL	88.4	85.3
60	JITNet-Iris	90	87.2
60	LD-MTN	93.2	90.6
80	MTL-IrisNet	88.3	84.3
80	Dual-Attentive MTL	89.3	86.2
80	JITNet-Iris	91	88
80	LD-MTN	94.1	91.7
100	MTL-IrisNet	88.1	86.5
100	Dual-Attentive MTL	89.5	88
100	JITNet-Iris	90.4	89.9

Table 18: Equal Error Rate (EER, %) over Epochs

Epochs	Method	CASIA-IrisV4	ND-IRIS-0405
20	MTL-IrisNet	6.1	6.9
20	Dual-Attentive MTL	5.4	6.2
20	JITNet-Iris	4.6	5.4
20	LD-MTN	3.9	4.8
40	MTL-IrisNet	5.3	5.9
40	Dual-Attentive MTL	4.7	5.4
40	JITNet-Iris	3.9	4.8
40	LD-MTN	2.8	3.5
60	MTL-IrisNet	4.5	5.2
60	Dual-Attentive MTL	3.9	4.7
60	JITNet-Iris	3.1	4.1
60	LD-MTN	2.1	2.9
80	MTL-IrisNet	4	4.8
80	Dual-Attentive MTL	3.3	4
80	JITNet-Iris	2.5	3.4
80	LD-MTN	1.9	2.2
100	MTL-IrisNet	4.3	4.6
100	Dual-Attentive MTL	3.4	3.8
100	JITNet-Iris	2.8	2.9

100	LD-MTN	94.1	92.5	100	LD-MTN	1.7	1.9
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Table 19: Inference Time (ms/sample) over Epochs

Epochs	Method	CASIA-IrisV4	ND-IRIS-0405
20	MTL-IrisNet	42.8	43.1
20	Dual-Attentive MTL	40.1	40.3
20	JITNet-Iris	36.7	36.9
20	LD-MTN	26.3	26.7
40	MTL-IrisNet	42.6	43
40	Dual-Attentive MTL	40	40.1
40	JITNet-Iris	36.5	36.8
40	LD-MTN	26	26.4
60	MTL-IrisNet	42.3	42.9
60	Dual-Attentive MTL	39.9	40
60	JITNet-Iris	36.4	36.7
60	LD-MTN	25.8	26.2
80	MTL-IrisNet	42.1	42.7
80	Dual-Attentive MTL	39.7	39.8
80	JITNet-Iris	36.3	36.6
80	LD-MTN	25.5	26
100	MTL-IrisNet	42.2	42
100	Dual-Attentive MTL	39.5	39.3
100	JITNet-Iris	34.5	34.7
100	LD-MTN	25.4	25.8

5. Conclusion

This study presents LD-MTN, a novel Layer-Divergent Multi-Task Network for joint iris segmentation and recognition. By allowing task-specific branches to diverge at optimal depths via attention-based dynamic selection, LD-MTN overcomes the limitations of uniformly shared architectures in conventional MTL models. Extensive experimentation on CASIA-IrisV4 and ND-IRIS-0405 datasets validates the model's superiority. The proposed framework outperforms strong baselines like JITNet-Iris and Dual-Attentive MTL, achieving improvements of up to 1.9% in recognition accuracy, 4.1% in IoU, 3.7% in segmentation F1 score, and a 39.3% reduction in EER. Furthermore, it operates at over 25% faster inference speed, making it highly suitable for real-time applications. LD-MTN's intelligent feature sharing strategy ensures that low-level pixel details are prioritized for segmentation, while deep embedding are reserved for identity recognition. The integrated loss optimization and modular design make it scalable and adaptable to other biometric or medical imaging tasks.

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